

Phase : Project developement phase

Date	25 september 2026
Team ID	SWTID-2026-7377
Project name	script-controlled-ACL
Maximum marks	4

The Project Development Phase (or Execution Phase) is where the design blueprints are transformed into functional software through coding, configuration, integration, and continuous testing.

1. Key Objectives

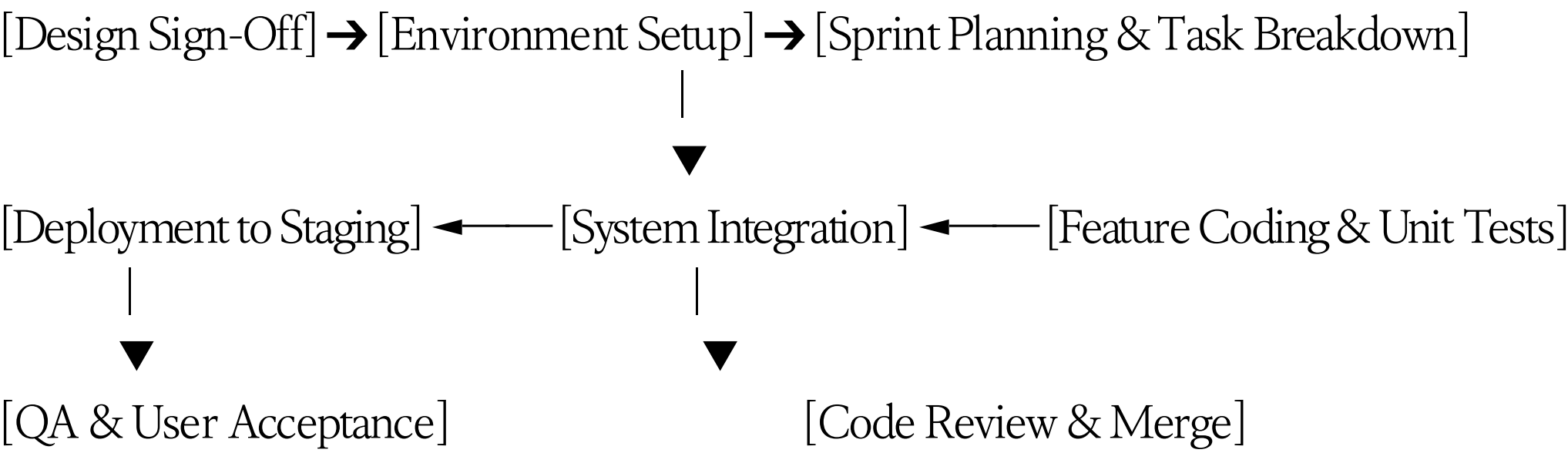
- Construct the Product: Write clean, maintainable code and configure system components according to the Design Document.
- Iterative Delivery: Build features in incremental units (sprints/modules) to allow early validation and feedback.
- Maintain Code Quality: Enforce coding standards, perform peer reviews, and run automated testing.
- Integrate Systems: Connect frontend interfaces, backend services, databases, and third-party APIs.

2. Core Activities & Deliverables

Category	Key Activities	Output Deliverables
Environment Setup	Provisioning development, testing, and staging environments; configuring CI/CD pipelines.	Configured Dev/Test Infrastructure, Build Pipelines
Code Execution	Front-end/back-end coding, database migration scripts, system configuration, unit testing.	Source Code, System Build Artifacts, Database Schemas

Quality Assurance	Peer code reviews, continuous integration execution, bug identification and remediation.	Code Review Checklists, Passing Unit Test Logs
Technical Documentation	Writing API specifications, code inline documentation, system configuration guides.	API Docs, Technical Manuals, Release Notes

3. High-Level Process Workflow



1. Input: Technical Design Document (TDD), Architecture Specs, and Backlog/User Stories.
2. Environment & Pipeline Setup: Establish source code repositories, feature branching strategies, and CI/CD triggers.
3. Sprint Execution: Developers pick up user stories, code solutions, write unit tests, and

submit Pull Requests (PRs).

4. Code Review & Static Analysis: Automated tools and peer reviewers verify code against quality and security guidelines.
5. Output: Integrated, tested build ready for formal Quality Assurance (QA) and User Acceptance Testing (UAT).

4. Key Best Practices

- Version Control Discipline: Enforce a structured branching strategy (e.g., GitFlow, Trunk-Based Development) with protected main branches.
- Continuous Integration (CI): Run automated builds and unit test suites on every pull request to catch regressions early.
- Defect Management: Track bugs using prioritized backlogs (e.g., Jira) with standardized severity labels (Blocker, Critical, Major, Minor).

Are you looking to tailor this phase for a specific framework (like Agile/Scrum vs. Waterfall) or focus on developer environment tooling and CI/CD pipelines?